

Problem 1

- Null model:

$$L(w) = \sum_{j=1}^n y_j(w_0 + \sum_{i=1}^p w_i x_{ji}) - \ln(1 + \exp(w_0 + \sum_{i=1}^p w_i x_{ji}))$$

According to the notes, we basically do an Odds ratio decision rule:

dataset	best eta	train misclass	test misclass
Gisette	0.200000	0.000000	0.020000
Madelon	1.000000	0.259500	0.426667
Dexter	0.200000	0.000000	0.140000

Table 1: Best learning rate η , training and test misclassification error, Logistic Regression.

Problem 2

- asdf
- aghjk